Vision Career

Comprehensive Product, Technical, and Implementation Documentation

Professional master document for final project review, implementation, demo preparation, and future scaling

Project	Vision Career
Document purpose	Single-source reference for architecture, flows, AI logic, phases, data model, integrity rules, and implementation scope
Primary platform	Flutter mobile application
Core AI integrations	Gemini for recommendation / quiz / generation; Vertex AI Search for learning resources
Current product model	Phase-based guided path system: Phase 0 -> Phase 1 -> Phase 2 -> Phase 3

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1. Executive Summary

Vision Career is a mobile-first guided education and career-planning system that helps university students move from uncertainty to job readiness through a structured, phase-based path. The product combines deterministic prerequisite logic with AI-assisted recommendation, assessment generation, and final-career preparation.

The system is intentionally designed so that large language models add intelligence without taking control. Models can rank, summarize, and generate suggestions, while the application remains responsible for validation, navigation, persistence, prerequisite enforcement, progress, and user safety.

At a product level, the experience is built around four major stages:

- Phase 0 - guidance and discovery before entering a path.
- Phase 1 - foundation subjects for the selected college or study area.
- Phase 2 - specialization subjects for the selected major.
- Phase 3 - final career-readiness topics generated from the user's completed path and target jobs.

Why this system matters

- Students often choose majors and courses with weak guidance, incomplete market awareness, and no clear learning sequence.
- Traditional advising gives information, but not an interactive progression engine.
- Online learning platforms offer many resources, but no integrated academic path, assessment loop, or career-finish layer.
- Vision Career connects guidance, learning order, assessment, integrity, and final readiness in one coherent product.

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2. Product Vision and Problem Statement

2.1 Product vision

Vision Career aims to become a smart academic navigation platform that does more than display majors. It should help a student discover where they fit, understand what to study next, prove they learned it, and complete a final bridge from specialization knowledge to career readiness.

2.2 Primary user problem

A typical student does not only need a list of subjects. They need a system that answers four deeper questions:

- Where do I fit?
- What should I study first, second, and later?
- How do I prove I understood each step?
- What do I still need before I am ready for work?

2.3 Product answer

Vision Career answers these questions with a combination of guided onboarding, prerequisite trees, node-level resources, quizzes with progression rules, and a final AI-assisted readiness stage.

2.4 Design philosophy

- Impressive for demos, but technically safe in execution.
- AI-assisted, not AI-chaotic.
- User chooses; the app never auto-commits a path without explicit confirmation.
- Nodes are visual, interactive, and progression-based rather than passive text lists.
- Every major system must be recoverable, explainable, and testable.

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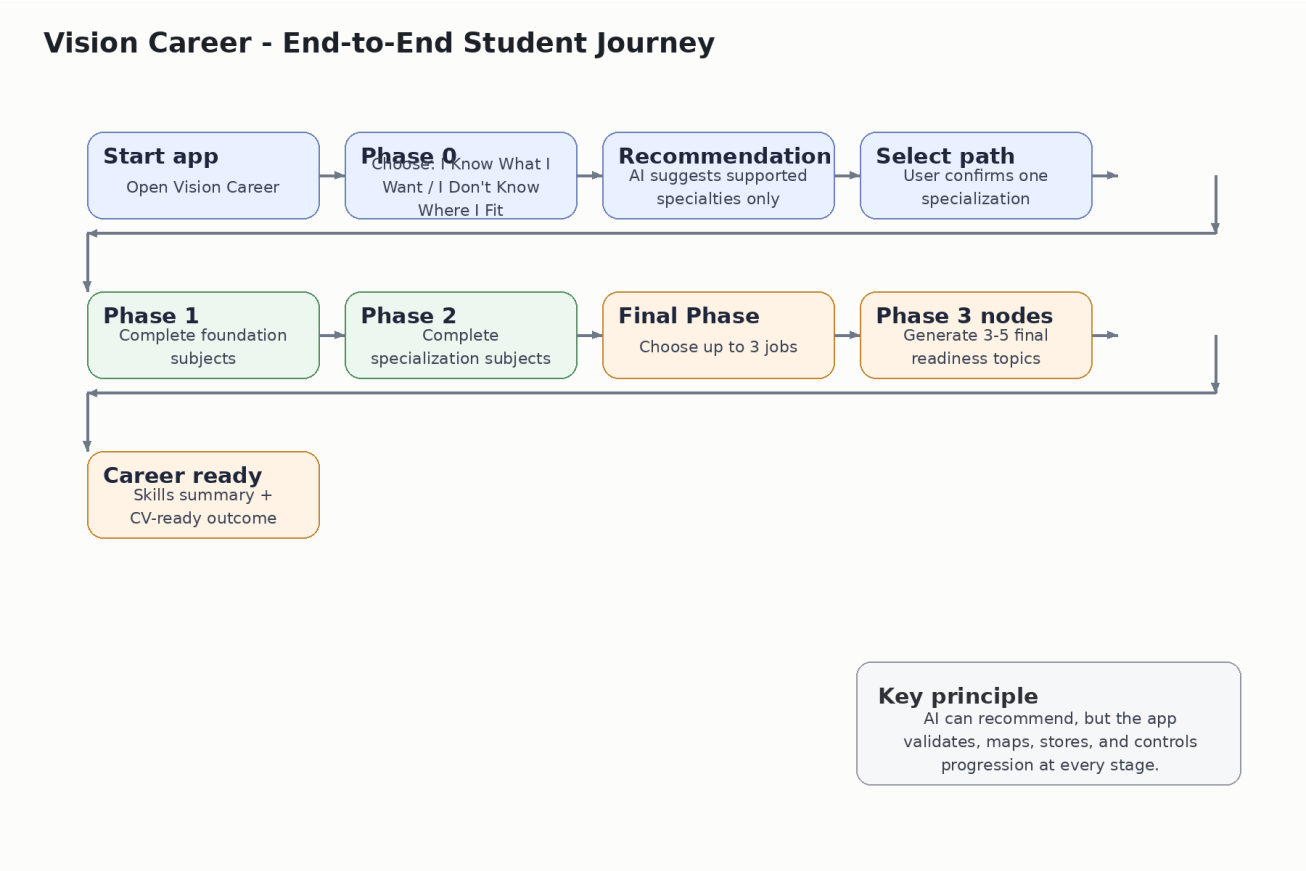
3. Core Experience and Student Journey

3.1 End-to-end student journey



The student journey begins before path generation. Instead of forcing early manual selection, the system can guide the student into a suitable specialization and then keep them inside one persistent progression loop from discovery to job readiness.

3.2 Journey narrative

1. The student opens the app and enters Phase 0.
2. The student either describes what they already want or completes a guided discovery flow.
3. The app shows only supported specializations and requires the user to confirm one.
4. The tree opens with the correct local dataset and prerequisite graph.

5. The student explores node details, learning resources, gained skills, and requirements.
6. Completion is controlled by assessment rules; passing unlocks the next valid nodes.
7. Once Phase 1 and Phase 2 are complete, FINAL PHASE becomes available.
8. The student selects up to three possible jobs, receives a generated final mini-path, and completes the last readiness nodes.
9. The final output is a clearer job target, stronger skills summary, and a more CV-ready profile.

3.3 Product value for the student

Need	Typical student pain	Vision Career response
Direction	Unsure what major or specialty fits	Phase 0 recommendation flows
Order	Does not know subject sequence	Prerequisite-driven tree
Confidence	Cannot tell if they actually learned	Quiz gating and progression
Resources	Too many random learning links	Ranked resource retrieval
Career finish	Unsure what comes after specialization	Phase 3 career-readiness layer

4. Vision Career Phase Architecture

4.1 Phase overview

Phase	Purpose	Main input	Main output
Phase 0	Guidance and recommendation	User intent / discovery data	Selected supported specialty
Phase 1	Foundation learning	College / specialization mapping	Completed core subjects
Phase 2	Specialization progression	Phase 1 completion + prerequisites	Completed specialization subjects
Phase 3	Career readiness completion	Completed path + selected jobs	Final readiness nodes and skills summary

4.2 Phase 0 summary

Phase 0 replaces the older college-selection and specialization-selection entry route. It is the intelligent front door of the app. The goal is not to let the AI invent majors; the goal is to help the user choose from paths the app already supports.

4.3 Phase 1 summary

Phase 1 contains the foundational subjects that establish college-level or discipline-level readiness. These nodes should unlock in dependency-safe order and form the base of later specialization.

4.4 Phase 2 summary

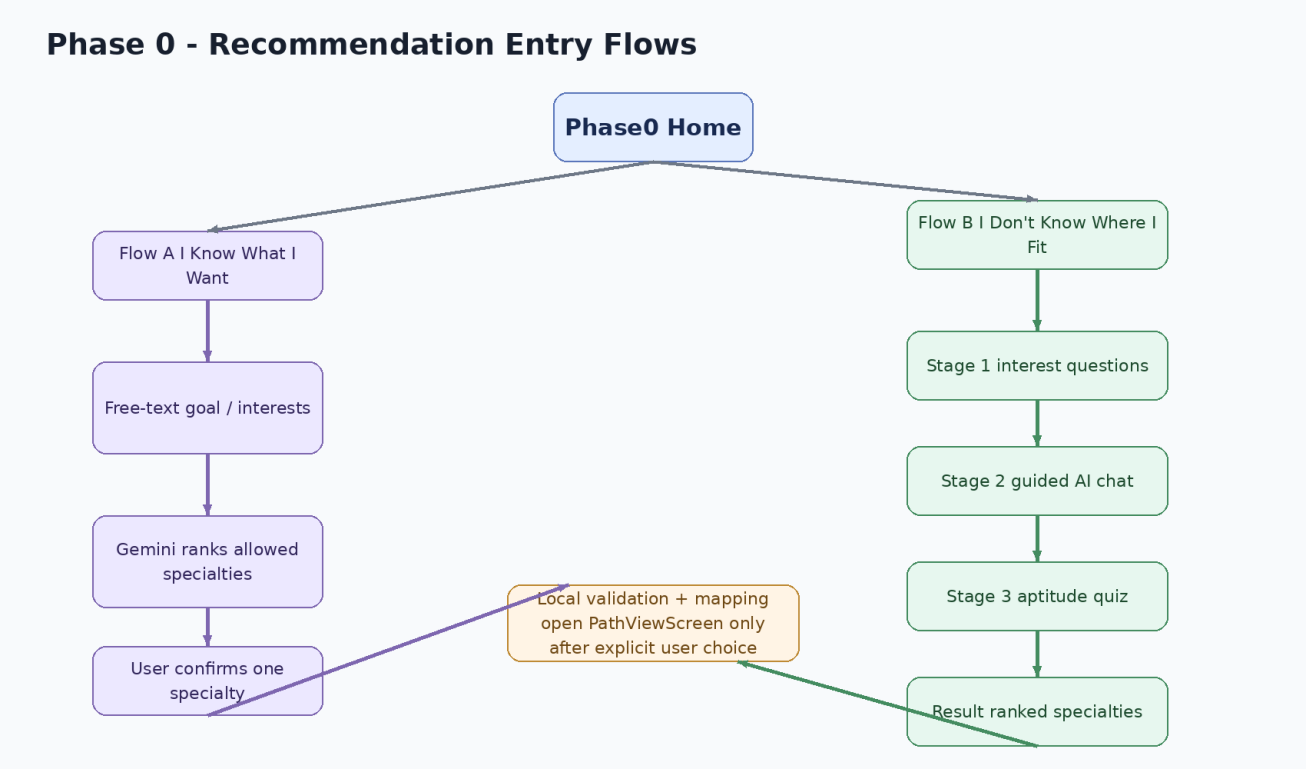
Phase 2 contains specialization-specific subjects. These are the deeper technical or domain nodes that move the user toward a defined study area.

4.5 Phase 3 summary

Phase 3 is the final transformation layer. After the learner completes the academic path, the system asks what job targets matter to them, then generates a small set of final topics that close practical readiness gaps.

5. Detailed User Flows

5.1 Phase 0 flow architecture



5.2 Flow A - I Know What I Want

1. User opens a free-text screen and describes what they want to study, build, or become.
2. The app sends the description together with the allowed specialty registry.
3. Gemini returns ranked specialties in strict JSON format.
4. The app filters invalid outputs, keeps only locally supported options, and displays them.
5. The user explicitly taps one recommendation to continue.
6. The app maps the specialty key to the local dataset and opens the tree.

5.3 Flow B - I Don't Know Where I Fit

7. Stage 1 collects lightweight interest questions using quick multiple-choice cards.
8. Stage 2 runs a bounded guided AI chat to clarify motivation, school preferences, and work style.
9. Stage 3 runs a soft aptitude quiz using simple math, logic, and reading signals.
10. The app sends summarized structured signals plus the allowed registry to Gemini.
11. Gemini returns ranked specialties from the allowed list only.
12. The app validates, displays, and waits for user confirmation before navigation.

5.4 Tree interaction flow

13. PathViewScreen displays the subject graph as an explorable tree.
14. Locked nodes remain visible and clickable so the student can inspect descriptions and requirements.

15. Unlocked nodes use a clear active visual state.
16. Completed nodes use a distinct completion state.
17. Long-press or controlled action opens the completion path, which is quiz-gated when required.
18. Dependency rules prevent invalid progression and should also prevent unsafe reverse actions that break consistency.

5.5 FINAL PHASE flow

19. The FINAL PHASE button becomes available only after Phase 1 and Phase 2 are complete.
20. The app asks Gemini for a set of possible job roles using completed-path context.
21. The user can select up to three jobs.
22. The app asks Gemini for 3-5 final topics based on selected jobs, specialization, and completed subjects.
23. The app converts generated topics into new nodes that behave like normal learning nodes.
24. These nodes fetch resources, require quizzes, persist locally, and cannot be regenerated once committed.

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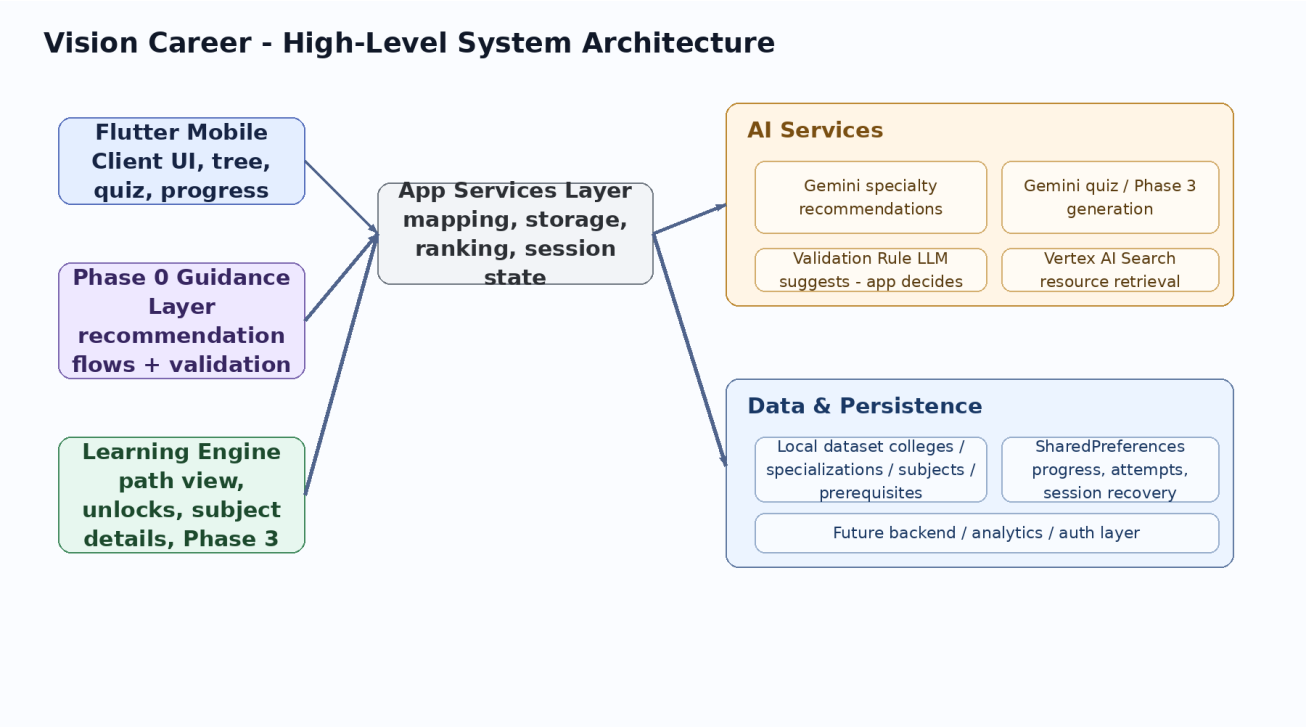
6. Functional Modules

Module	Responsibility
Recommendation engine	Handles Phase 0 prompting, response parsing, local filtering, and safe retry behavior.
Mapping layer	Maps specialty keys to internal college / specialization / dataset identifiers.
Tree engine	Loads nodes, prerequisites, status, and unlock state for the chosen path.
Subject details layer	Displays descriptions, gained skills, status, resources, and completion actions.
Quiz engine	Builds or evaluates assessments, computes pass/fail, and returns result states.
Integrity layer	Applies anti-cheat controls and logs meaningful session flags.
Resource retrieval layer	Queries Vertex AI Search and ranks results using platform priority rules.
Phase 3 builder	Converts generated career-preparation outputs into persistent path nodes.
Storage layer	Persists progress, attempts, session recovery, and generated Phase 3 state.

7. AI Systems and External Services

7.1 High-level architecture



7.2 AI responsibilities

- Gemini in Phase 0: rank supported specialties and explain fit.
- Gemini in quizzes: generate or assess question sets according to the subject context.
- Gemini in Phase 3: suggest job roles and generate final readiness topics.
- Vertex AI Search: return ranked learning resources for node topics.

7.3 Control boundary

A strict boundary should remain in place across the whole system: LLMs produce candidate outputs; the application validates and decides what becomes real product state.

7.4 Resource ranking rules

- Maximum four resources per node.
- Maximum two Coursera results.
- Platform priority: Coursera -> Udemy -> YouTube.
- The app should deduplicate and discard weak or irrelevant results.
- Topic generation and resource retrieval remain separate responsibilities.

7.5 Error handling expectations

- Malformed JSON should trigger parse-safe fallback and retry UI.
- Invented majors or unknown specialty keys must be dropped.
- Empty valid results must not silently continue; the user should receive a clear retry option.
- External API failures should not corrupt saved progress or route state.

8. Data Model and Persistence

8.1 Core data entities

Entity	Fields	Purpose
College	id, name	Top-level academic grouping
Specialization	id, college_id, key, title	Selectable path identity
Subject / node	id, title, phase, credits, description, skills	Learning unit shown in the tree
Prerequisite edge	subject_id, prerequisite_subject_id	Dependency relation for unlocking
User progress	user_id, subject_id, completed, score, attempts	Progress and result tracking
Phase 0 session	answers, summaries, chosen specialty	Recovery for unfinished recommendation flows
Phase 3 state	selected_jobs, generated_nodes, completion states	One-time final-phase persistence

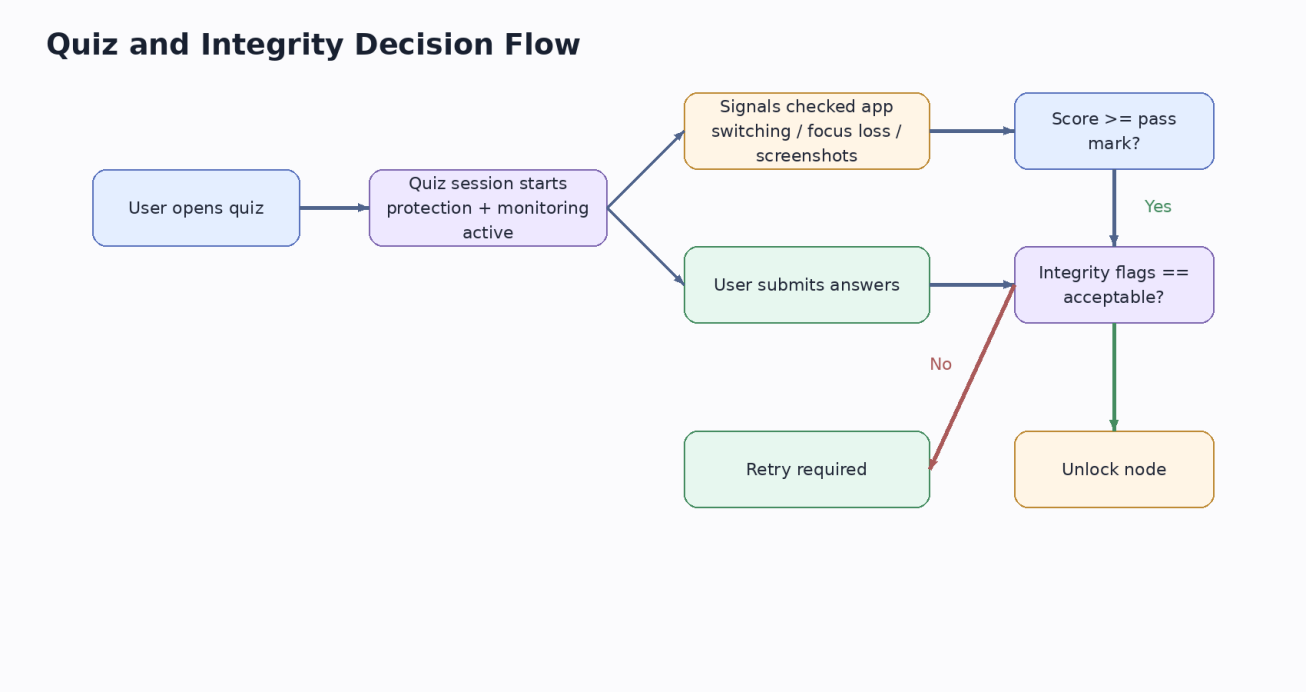
8.2 Persistence rules

- Normal path progress should persist independently from temporary Phase 0 session data.
- Phase 0 should support mid-flow recovery after app close or interruption.
- Phase 3 generated nodes must be stored once committed and must not be regenerated casually.
- Attempt limits and integrity-related counters should be stored only as far as they are actually used in product logic.

8.3 Recommended specialty registry

The application should maintain a single source of truth for all specialties Gemini is allowed to suggest. This registry protects the product from model drift and invented outputs.

9. Quiz, Integrity, and Unlock Logic



9.1 Quiz role in progression

Quizzes are not decorative. They are a formal progression gate. Their role is to verify that the student is ready to unlock the next nodes.

9.2 Assessment rules

- A node should only complete after the required quiz condition is satisfied.
- Passing threshold should be configurable by phase or quiz type.
- Integrity flags must be checked before accepting a passing result.
- Failure should trigger a clear retry path instead of ambiguous state.

9.3 Integrity controls

- App switching and focus loss detection during quiz sessions.
- Screenshot protection and anti-copy behavior during assessments.
- Session flagging for abnormal behavior.
- Minimal and useful logging only; avoid storing noise that has no product value.

9.4 Unlock logic

- Locked nodes remain visible but unavailable for completion.
- Unlocked nodes appear only when prerequisites are satisfied.
- Completed nodes should not be allowed to revert in a way that breaks downstream dependency consistency.
- The UI should clearly communicate why a node is locked and what is required next.

10. Mobile Application Architecture

10.1 Architectural style

The current product is a Flutter mobile application with local-first behavior and optional cloud-backed services. It mixes deterministic path logic with selective AI calls and should be treated as a modular client application rather than a thin UI shell.

10.2 Major layers

- Presentation: screens, widgets, navigation, interaction states.
- Application / services: mapping, prompting, storage, ranking, unlock evaluation.
- Data: local JSON or structured data files for colleges, specializations, subjects, and prerequisites.
- External integrations: Gemini and Vertex AI Search.

10.3 Proposed runtime flow

25. UI action occurs.
26. Screen calls service or repository layer.
27. Service validates local state and external contract input.
28. External AI or search response returns.
29. Service parses and filters output.
30. UI updates only after a safe local product state is formed.

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11. File Structure and Responsibilities

11.1 High-level project tree

```
vision_career_mobile/
|- lib/
|   |- app/
|   |- core/
|   |   |- constants/
|   |   |- services/
|   |   |- models/
|   |   |- widgets/
|   |- data/
|   |   |- datasources/
|   |   |- repositories/
|   |   |- models/
|   |- features/
|   |   |- phase0/
|   |   |- path_view/
|   |   |- subject_details/
|   |   |- progress/
|   |   |- career/        # final phase
|   |- shared/
|- assets/
   |- data/
   |- images/
```

```
|- icons/
|- json/
```

11.2 Phase 0 files

- phase0_home_screen.dart - two-option landing screen.
- know_what_i_want_screen.dart - free-text collection.
- specialty_recommendation_screen.dart - recommendation display and confirmation.
- fit_discovery_questions_screen.dart - stage 1 interest questions.
- fit_discovery_chat_screen.dart - stage 2 guided AI chat.
- fit_discovery_quiz_screen.dart - stage 3 aptitude quiz.
- fit_discovery_result_screen.dart - ranked results and final handoff.
- phase0_gemini_service.dart - prompt building, parse logic, validation.
- phase0_mapping_service.dart - specialty key to dataset mapping.
- phase0_session_storage_service.dart - recoverable temporary state.

11.3 Final phase files

- career_selection_screen.dart - job-role selection UI.
- career_analysis_screen.dart - generated topic result flow.
- career_llm_service.dart - Phase 3 prompting.
- career_phase3_builder.dart - convert generated topics into nodes.
- career_storage_service.dart - persistent Phase 3 state.

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12. User Interface and Interaction Model

12.1 Tree design principles

- Tree should feel explorable and visual, not like a static list.
- Locked, unlocked, and completed states should have clearly different treatments.
- Wide trees should support panning or zoom-friendly interaction rather than forcing cramped layouts.
- Node details should remain accessible even when a node is locked.

12.2 Recommended state language

State	Meaning	Recommended visual
Locked	Prerequisites not yet satisfied	Muted gray / shadowed
Unlocked	Available to attempt	Active blue
Completed	Requirement satisfied	Gold / highlighted success state

12.3 Node content expectations

- Title
- Description
- Skills gained
- Prerequisites / unlock requirements
- Learning resources

- Completion action and status

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13. Non-Functional Requirements

Reliability

- The app should preserve progress correctly after closes and restarts.
- AI failures should degrade gracefully without corrupting path state.

Performance

- Tree rendering should remain responsive on mobile devices.
- External calls should display clear loading states and retry actions.

Usability

- Flows must feel short, understandable, and visual.
- Error states must be user-readable, not developer-style messages.

Maintainability

- Service responsibilities should stay separated.
- AI contracts should be explicit and versionable.

Scalability

- The system should support more colleges and specializations without rewriting the entry architecture.
- The recommendation registry should make multi-college expansion safer.

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14. Risks, Constraints, and Mitigations

Risk	Why it matters	Mitigation
LLM invents unsupported majors	Can break navigation and trust	Use allowed specialty registry and local filtering
Malformed AI output	Can crash or confuse flows	Strict JSON contracts plus retry state
Feature creep late in project	Can reduce stability before demo	Freeze architecture and focus on polish
Weak data quality	Hurts recommendation and path credibility	Clean priority specializations and validate datasets
Over-logging integrity data	Adds complexity without value	Store only signals actually used by product logic